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Where Minds Pollinate

Class 12
Physics

Chapter 24
Printed Notes

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EARTH'S CLIMATE SYSTEM

- It is defined as a complex system with five interacting components; The atmosphere, hydrosphere, cryosphere, lithosphere and biosphere.
- It is influenced by the factors including the sun, earth's position in space and human made factors such as greenhouse gas emissions

1. The Atmosphere (Air)

- The atmosphere is the layer of air that surrounds the Earth.
- It is made of different gases and protects us by giving oxygen, controlling temperature, and blocking harmful rays from the Sun.
- It primarily consists of 78% nitrogen, 21% oxygen and 0.9% argon.

Role of Atmosphere

- Shielding: Atmosphere protects life on Earth by shielding it from incoming ultraviolet radiations
- Insulation: It insulates the planets, acting as an insulated layer that prevents extreme temperature variations.
- Greenhouse effect: The atmosphere traps some of the Sun's heat using gases like carbon dioxide and water vapor. This trapped heat keeps the Earth warm enough to live on.

Layers of Atmosphere

Atmosphere is divided into 5 layers:

- **Troposphere**: The lowest most player is known as troposphere It extends from the ground to about 16 km. In this layer weather occurs.
- **Stratosphere**: Stratosphere starts above the troposphere, from 16 to 50 km. This layer contains the ozone layer which prevents from harmful ultraviolet radiations coming from the sun.
- **Mesosphere**: After stratosphere, mesosphere extends from 50 to 136 km. It is the layer where meteors burn out.
- **Thermosphere**: Thermosphere extends from 136 to about 600 km. In this layer Auroras occurs due to the solar radiations.
- **Exosphere**: Exosphere is the outermost layer that extends from 600 km and merge into the space. It is the layer in which satellites orbits.

2. The Hydrosphere (Water)

- The hydrosphere is all the water present on Earth, in any form; liquid (oceans, rivers), solid (ice, snow), or gas (water vapor in the air).
- Earth's surface contains approximately 70% water due to this large amount of water, it show it's "blue glow" appearance.

Role of Hydrosphere

- Stores Heat: Oceans absorb and store a lot of the Sun's heat, helping keep Earth's temperature steady.
- Moves Heat Around: Ocean currents carry warm and cold water to different places, affecting weather and climate.
- Water for Rainfall: Water evaporates from oceans and forms clouds, which bring rain and snow.

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- Controls Temperature: Water heats up and cools down slowly, helping prevent sudden changes in Earth's temperature.

Water Balance Equation:- The water balance equation is a fundamental tool used to understand the moment and distribution of water in the hydrosphere our time.

$$\Delta S = P - E - Q$$

Where;

- P = Precipitation: Water that falls from the sky such as rain, snow, sleet, and hail. It is the main input of water.
- Q = Runoff (Discharge): Water that flows on the Earth's surface into rivers, lakes, and oceans. It is an output of water from land.
- E = Evapotranspiration: Loss of water to the atmosphere through evaporation (from water bodies and soil) and transpiration (from plants).
- ΔS = Change in Storage: Increase or decrease of water stored in soil, groundwater, lakes, glaciers, and reservoirs.

3. The Cryosphere (Ice)

- The cryosphere is the part of the Earth where water is frozen due to low temperature.
- Found mainly in polar regions (Arctic and Antarctic) and high mountains.

Role of Cryosphere

- Reflects Sunlight: Ice and snow reflect sunlight (albedo effect), keeping Earth cooler.
- Stores Fresh Water: Holds large amounts of Earth's fresh water.
- Regulates Sea Level: Melting ice increases sea levels; freezing lowers them.
- Influences Weather and Ocean Currents: Cold regions affect air and ocean circulation, impacting climate worldwide.

Components of Cryosphere

There are three main components of cryosphere:

Snow

- Frozen water falling from the sky that accumulates on the ground.
- Exists in cold regions and high mountains where temperatures are low.
- Role in climate: reflects sunlight (keeps Earth cool), stores fresh water, affects weather patterns.

Ice

- Includes glaciers, ice sheets, and sea ice (large masses of frozen water).
- Exists in polar regions and high mountains where ice accumulates over long periods.
- Role in climate: reflects solar energy, regulates sea levels, influences ocean currents and air temperature.

Permafrost

- Soil, rocks, or sediment that stays frozen all year round.
- Exists in polar and subpolar regions with permanently cold ground.
- Role in climate: stores carbon and methane, affects soil and ecosystems, keeps local areas cold.

4. The Lithosphere (Earth's Upper Rocky Layer)

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- The lithosphere is outermost solid shell, consisting of the crust and the uppermost part of the mantle, composed of various rocks and minerals.
- It forms the basis for continents, ocean basins and landforms by interacting with others components of earth's climate system.

Major Processes of Lithosphere (Components)

Land Surface:

- Provides the physical surface where climate processes occur.
- Different land types (deserts, forests, plains, mountains) influence climate in different ways.
- Deserts heat up quickly, forests add moisture, and mountains change wind and rainfall patterns.

Carbon Cycle:

- Rocks weather and erode, releasing carbon dioxide into the atmosphere.
- Some rocks like limestone store carbon for millions of years.
- Acts as a long-term carbon sink that helps balance Earth's climate over geological time.

Tectonic Activity:

- Movement of tectonic plates shapes Earth's surface over long time scales.
- Formation of mountain ranges affects atmospheric circulation and precipitation.
- Slow movement of continents influences global climate patterns.

Volcanic Activity:

- Volcanic eruptions release gases and particles into the atmosphere.
- Ash and particles can block sunlight and temporarily cool the climate.
- Gases like CO₂ released during eruptions can impact long-term climate.

5. The Biosphere (Living Things)

- The biosphere is the thin layer of life on Earth. It spreads from a few kilometers into the air to the deep sea floor.
- It includes all living things (plants, animals, fungi, microorganisms) and the non-living factors they depend on for energy and nutrients.
- Living organisms interact with the environment and affect Earth's climate system through processes like breathing, growing, and decomposing.

Role of Biosphere

- **Climate Regulation:** Plays a crucial role in regulating Earth's temperature by absorbing sunlight, releasing moisture, and keeping air gases in balance.
- **Carbon Cycle:** Acts as an integral reservoir in the carbon cycle as plants absorb carbon dioxide (CO₂) and reduce it in the atmosphere.
- **Energy Flow:** Plays an essential role in energy flow because plants capture solar energy and pass it through the food chain.
- **Water Cycle:** Helps drive the water cycle by releasing water vapor through transpiration, which supports cloud formation and rainfall.
- **Nutrient Cycle:** Acts as a key system in nutrient cycling by breaking down and recycling nutrients that support life and influence climate.

Ocean Currents and Winds

Ocean currents and wind patterns are the interconnected phenomena that play crucial roles in regulating Earth's climate and distributing heat globally.

Ocean Currents

- Ocean currents are large movements of seawater that flow like rivers within the ocean, carrying warm or cold water across the globe.
- They help regulate Earth's climate by moving heat from warm areas to cooler areas, keeping temperatures balanced.
- They also influence weather and marine life, as warm and cold currents affect rainfall, storms, and the distribution of sea animals and plants.
- They are categorized into two types:
 - (i) Surface Ocean Currents
 - (ii) Deep Ocean Currents

1) Surface Ocean Currents

- Surface ocean currents are movements of water in the upper, shallow layer of the ocean.
- They carry warm and cold water across long distances, helping to shape climate and weather patterns.

Factors:

- **Wind** plays a major role by pushing the surface water and setting it into motion.
- **Subtropical gyres** (large rotating systems) help guide these currents, moving warm water toward the poles and collecting floating materials like plastic.

2) Deep Ocean Currents

- Deep ocean currents are large movements of water found below 500 meters in the ocean.
- They form due to differences in water temperature and density and help move cold and warm water around the globe, affecting climate over long time scales.

Factors:

- **Warm Water Moving Northward:** Warm surface water travels toward the poles, cools down, and sinks, helping drive deep ocean circulation.
- **Density Differences:** Water becomes denser when it is colder and saltier, causing it to sink to the ocean depths. This sinking creates deep circulation, moving cold, dense water along the ocean floor toward the equator. This process, known as the global conveyor belt, distributes heat and nutrients worldwide and influences Earth's climate.

Coriolis Effect

Definition: The Coriolis effect is the deflection of moving water or air caused by Earth's rotation.

Cause: Earth rotates from west to east, which makes moving water bend to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.

Effect on Ocean Currents: This deflection creates loops in ocean gyres — clockwise in the Northern Hemisphere, counterclockwise in the Southern Hemisphere.

Role in Climate: It guides surface ocean currents, distributing heat and influencing regional climate and weather patterns.

Earth's Angular Momentum and Wind Patterns**Angular Momentum of Earth:**

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- Angular momentum is the tendency of a rotating body (like Earth) to keep spinning unless acted upon by an external force.
- Earth's rotation gives the atmosphere and oceans a conserved angular momentum, which affects the motion of air and water.
- This rotation is the fundamental reason why the Coriolis effect exists.
- **Coriolis Effect:** The Coriolis effect is the apparent deflection of moving objects (like air or ocean currents) caused by Earth's rotation.
- It is a direct consequence of Earth's angular momentum, as moving air tries to conserve momentum while Earth rotates beneath it.
- Effect by hemisphere:
 - Northern Hemisphere: deflection to the right
 - Southern Hemisphere: deflection to the left
- **Coriolis Force:** The Coriolis force is the actual force acting on a moving mass due to Earth's rotation.
- Formula: $F = 2mv\omega\sin\theta$
Where: F = Coriolis force, m = mass of moving air/water, v = velocity of the air/water, ω = angular velocity of Earth and θ = latitude.
- Variation with latitude: Maximum at the poles ($\theta=90^\circ$), Zero at the equator ($\theta=0^\circ$).
- **Influence of moment of inertia:** The distribution of Earth's mass affects how fast regions rotate, influencing wind speed and deflection.
- **Influence on wind speed:** Faster winds experience a stronger Coriolis force, resulting in more pronounced deflection.

Formation of Atmospheric Circulation Cells

Earth's rotation and the Coriolis force divide the atmosphere into three main circulation cells per hemisphere:

1. Hadley Cells (0° – 30°): Warm air rises at the equator, moves poleward, cools, and sinks at $\sim 30^\circ$ latitude.
2. Ferrell Cells (30° – 60°): Air circulates between Hadley and Polar cells; winds move opposite to Hadley and Polar cells.
3. Polar Cells (60° – 90°): Cold air sinks at the poles, flows toward $\sim 60^\circ$ latitude, and rises to complete the loop.

Global Influence on Climate and Weather

- Hadley cells create tropical rainforests and deserts ($\sim 30^\circ$ latitude).
- Ferrel and Polar cells influence temperate and polar climates, including storms and precipitation.

Wind Patterns:

- Wind is the movement of air from high-pressure to low-pressure areas.
- Wind patterns are shaped by Earth's rotation, pressure differences, and friction.
- **Geostrophic wind** is a wind that flows parallel to isobars due to a balance between two main forces.

- **Isobars:** Lines on a map connecting points of equal pressure; the closer the isobars, the stronger the wind.
- **Forces Involved:**
 - a. Horizontal Pressure Gradient Force (PGF): Pushes air from high-pressure to low-pressure regions.
 - b. Coriolis Force: Deflects moving air due to Earth's rotation (right in Northern Hemisphere, left in Southern Hemisphere).

When these two forces balance exactly, the wind flows along the isobars, not directly from high to low pressure this is called **geostrophic balance**.

Influence on Climate and Weather

Heat Transport: Winds move warm air from tropics toward poles and cold air from poles toward the equator, balancing global temperatures.

Moisture Transport: Winds carry water vapor, influencing rainfall patterns, storms, and monsoons.

Pollutant Dispersion: Wind spreads air pollution, dust, and aerosols across regions.

Pollen Distribution: Winds aid in spreading pollen and seeds, affecting ecosystems and agriculture.

Climate Inertia

- Climate inertia is the resistance or slowness of the climate system to respond to changes in factors like greenhouse gas emissions, even when the cause of change is reduced or stopped.
- **Positive Feedback:** A process that amplifies climate change, e.g., melting ice reduces albedo, causing more warming and more ice melt.
- **Negative Feedback:** A process that reduces or slows climate change, e.g., more clouds reflect sunlight, slightly cooling the Earth.
- **Short-Term Inertia:** Climate responds slowly over months or years due to oceans' heat capacity and seasonal cycles.
- **Long-Term Inertia:** Climate changes very slowly over decades to centuries because ice sheets, deep ocean currents, and carbon storage in rocks take a long time to respond.
- **Importance for Adaptation:** Understanding climate inertia is essential for planning mitigation and adaptation strategies, as some effects of climate change cannot be reversed immediately.
- **Mitigation Steps:** Reduce greenhouse gas emissions, enhance carbon sinks (forests, soils), adopt renewable energy, and improve urban planning to cope with slow climate response.

Internal Variations of Climate

Internal variations are natural fluctuations within the climate system, caused by interactions between components like the atmosphere, oceans, and biosphere.

Examples:

- Solar Radiation: Natural changes in sunlight affect temperatures seasonally and over decades.
- Volcanic Eruptions: Large eruptions release ash and gases, temporarily cooling the climate.
- Ocean Oscillations: Phenomena like El Niño and La Niña change sea surface temperatures and rainfall patterns.
- Atmospheric Circulation Patterns: Shifts in wind belts and jet streams can redistribute heat and moisture, creating wet or dry periods.

External Forcing of Climate

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External forcing refers to factors outside the climate system that drive long-term changes.

Examples:

- Greenhouse Gas Concentration: Higher CO₂, methane, and nitrous oxide trap more heat, warming the Earth.
- Solar Variability: Changes in solar output over years or centuries affect global temperatures.
- Orbital Variations: Slight changes in Earth's orbit and tilt alter the distribution of sunlight and cause ice ages.
- Aerosols, Dust, and Pollution: Particles in the atmosphere reflect or absorb sunlight, affecting temperatures.

Earth Energy Budget

The Earth's energy budget is the balance between incoming energy from the Sun and outgoing energy from Earth. This balance determines Earth's overall temperature and climate.

Components of Earth's Energy Budget**Incoming Solar Radiation:**

Energy from the Sun reaches Earth in the form of shortwave radiation (visible light, UV, and some infrared). Part of it is reflected by clouds, ice, and surfaces back to space.

Atmospheric Absorption:

Some solar energy is absorbed by the atmosphere (clouds, water vapor, and gases), heating the air before it reaches the surface.

Outgoing Terrestrial Radiation:

Earth's surface absorbs solar energy and re-emits it as longwave infrared radiation. Some of this energy escapes to space, helping maintain Earth's temperature.

Greenhouse Effect

The greenhouse effect is the warming of Earth's surface caused by greenhouse gases trapping heat in the atmosphere.

Four Steps of the Greenhouse Effect:

1. Sunlight penetrates the atmosphere and reaches the Earth's surface.
2. The heated surface emits infrared radiation (longwave heat energy) back toward the atmosphere.
3. Greenhouse gases (CO₂, methane, water vapor) absorb some of this infrared radiation.
4. Greenhouse gases re-radiate part of this energy back toward Earth, keeping the surface warm enough to support life.

Without the Greenhouse Effect, Earth would be extremely cold, with an average temperature around -18°C, instead of the current ~15°C. Life as we know it would not survive without this natural warming process.

Solar Variability and Climate

Solar variability refers to changes in the Sun's energy output, which can affect Earth's climate over short and long periods.

Solar Cycles:

- The Sun undergoes approximately 11-year solar cycles, with fluctuations in sunspots and solar activity.

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- These cycles slightly change the amount of energy reaching Earth, causing minor warming or cooling effects.

Long-Term Trends:

- Over the past 40 years, observations show that the Sun's energy output has remained relatively stable.
- These long-term changes account for less than 0.1% of the warming observed since 1750.
- Therefore, solar variability cannot explain the rapid global warming seen in recent decades.

Human-Induced Warming:

- Modern global warming is primarily caused by human activities, especially emissions of greenhouse gases.
- The warming effect of these gases is about 270 times greater than any warming caused by changes in sunlight.
- This highlights that while the Sun influences climate slightly, human impact dominates current climate change.

Energy Imbalance

Earth receives unequal solar energy at different latitudes, causing an energy imbalance between the equator and the poles.

Energy Distribution

Equator: Receives more solar energy because sunlight strikes it directly. This heats the surface strongly, creating warm air that rises.

Poles: Receive less solar energy because sunlight comes at a slanted angle, spreading energy over a larger area. The surface remains cold, and air is dense and sinks.

Effects of Energy Imbalance

The difference in temperature and pressure between equator and poles creates pressure gradients that drive winds and circulation.

Atmospheric Circulation Patterns:

- Warm air rises at the equator → moves toward poles in the upper atmosphere → cools and sinks at ~30° and 60° latitudes.
- Creates Hadley Cells, Ferrel Cells, and Polar Cells.
- Trade winds, westerlies, and polar easterlies form due to Coriolis deflection.